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AF3316 Group Project Report

Event Study Analysis and Return Analysis of GLP-1 Breakthrough
Impact on US Medical Device Stocks

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1. Background

Obesity has evolved into a “system-wide market” rather than a standalone medical condition. It affects over one billion adults worldwide and generates persistent demand across multiple healthcare industries, including bariatric surgery, chronic disease management, cardiovascular/respiratory devices, and long-term monitoring and care. In this sense, obesity acts as a multi-sector demand engine that supports a broad medical ecosystem.

A major informational shock to this ecosystem occurred on **August 8, 2023**, when Novo Nordisk released the SELECT clinical trial results for Wegovy, a GLP-1 drug originally used for type-2 diabetes but increasingly recognized for meaningful weight-loss and broader metabolic and cardiovascular benefits. The SELECT results strengthened the market’s belief that GLP-1 therapies could shift obesity treatment away from traditional device- and procedure-based pathways toward pharmacological solutions. This is best interpreted as a “demand reallocation” event: part of the demand previously captured by surgery, medical devices, and long-term equipment may be redirected to medications.

Motivated by this demand reallocation mechanism, this report investigates whether the SELECT announcement produced a differential stock-market reaction between firms with business models directly exposed to obesity-related treatment demand and firms whose core demand is less dependent on obesity prevalence.

Specifically, we form a **treatment group** of healthcare firms whose revenues are closely linked to obesity-related treatment or monitoring demand and thus may face substitution risk from GLP-1 adoption, and a **control group** of healthcare firms whose core demand is comparatively independent of obesity rates.

This report consists of two sections: a short-term event study around the SELECT announcement and a long-term analysis of post-event performance and risk pricing, including a firm-size breakdown.

2. Methodology

This study combines event study analysis and longer-horizon risk–return analysis to evaluate how the release of Novo Nordisk’s Wegovy SELECT trial results affected healthcare stocks with different exposure to obesity-related demand. The event date is defined as August 8, 2023, and we examine trading days surrounding the announcement, including the August 4–10 window used in our analysis. Firms are classified into a treatment group whose demand is closely linked to obesity-related treatment and medical devices, and a control group whose core demand is less dependent on obesity rates. Using daily adjusted stock price data, we compute abnormal returns around the event and compare the short-term market reaction between the two groups.

To assess whether the event-driven repricing is consistent with longer-term performance and risk pricing, we further analyze changes in average returns and key risk–return indicators, including market beta, Security Market Line (SML) relationships, and risk-adjusted performance measures such as the Sharpe ratio. In addition, the treatment and control groups are each divided into large- and small-cap subgroups to form four portfolios, allowing us to examine whether firm size moderates the impact of the GLP-1 shock and to support subsequent investment strategy discussion.

The selected companies are demonstrated in the following table:

Treatment Group	Control Group
Intuitive Surgical (ISRG)	Johnson & Johnson (JNJ)
DexCom (DXCM)	Thermo Fisher Scientific (TMO)
ResMed (RMD)	Danaher (DHR)
Insulet (PODD)	Zoetis (ZTS)
Masimo (MASI)	IDEXX Laboratories (IDXX)
Medtronic (MDT)	Cooper Companies (COO)
Penumbra (PEN)	Align Technology (ALGN)
Inspire Medical Systems (INSP)	Boston Scientific (BSX)
Tandem Diabetes Care (TNDM)	Dentsply Sirona (XRAY)
Outset Medical (OM)	Haemonetics (HAE)
CONMED (CNMD)	Certara (CERT)
Edwards Lifesciences (EW)	Neo Genomics (NEO)
Merit Medical Systems (MMSI)	Castle Biosciences (CSTL)
LeMaitre Vascular (LMAT)	Guardant Health (GH)
AngioDynamics (ANGO)	STAAR Surgical (STAA)

Table 1: The firms selected in the Treatment Group and Control Group

3. Event Study Analysis

3.1 Treatment and Control Groups Data

Date	Treatment_AAR	Treatment_CAR	Control_AAR	Control_CAR	Diff_Treat_minus_Control	t_stat	p_value
2023/8/4	-0.009369961	-0.009369961	0.004657994	0.004657994	-0.014027955	-1.00494	0.3278443
2023/8/7	-0.014226922	-0.023463577	-0.011080606	-0.006474226	-0.003146316	-0.34273	0.7353326
2023/8/8	-0.031895039	-0.054610245	0.004516637	-0.001986831	-0.036411676	-4.18788	0.0003479
2023/8/9	-3.55E-06	-0.0546136	-0.002389049	-0.004371133	0.002385501	0.461021	0.6483567
2023/8/10	0.001662554	-0.053041844	-0.015283403	-0.019587731	0.016945957	2.415803	0.0232729
Full Window	-0.010766583	-0.053041844	-0.003915686	-0.019587731	-0.006850898	-1.56526	0.1201223

Table 2: Treatment and Control Groups Data

Table 2 summarizes the AAR, CAR, and the statistical comparison between the treatment and control groups. On the event day (August 8), the treatment group's AAR was -3.19%, while the control group's AAR was 0.45%. The difference of -3.64% is statistically significant ($p < 0.01$). This implies that the event had an immediate and significantly stronger negative effect on the treatment group. In contrast, over the full event window (August 4-10), the average difference in AAR is only -0.69%, with a p-value of 0.12 ($p > 0.10$), indicating that the event's impact was concentrated on the event day and did not accumulate over the full window.

3.2 Treatment Group

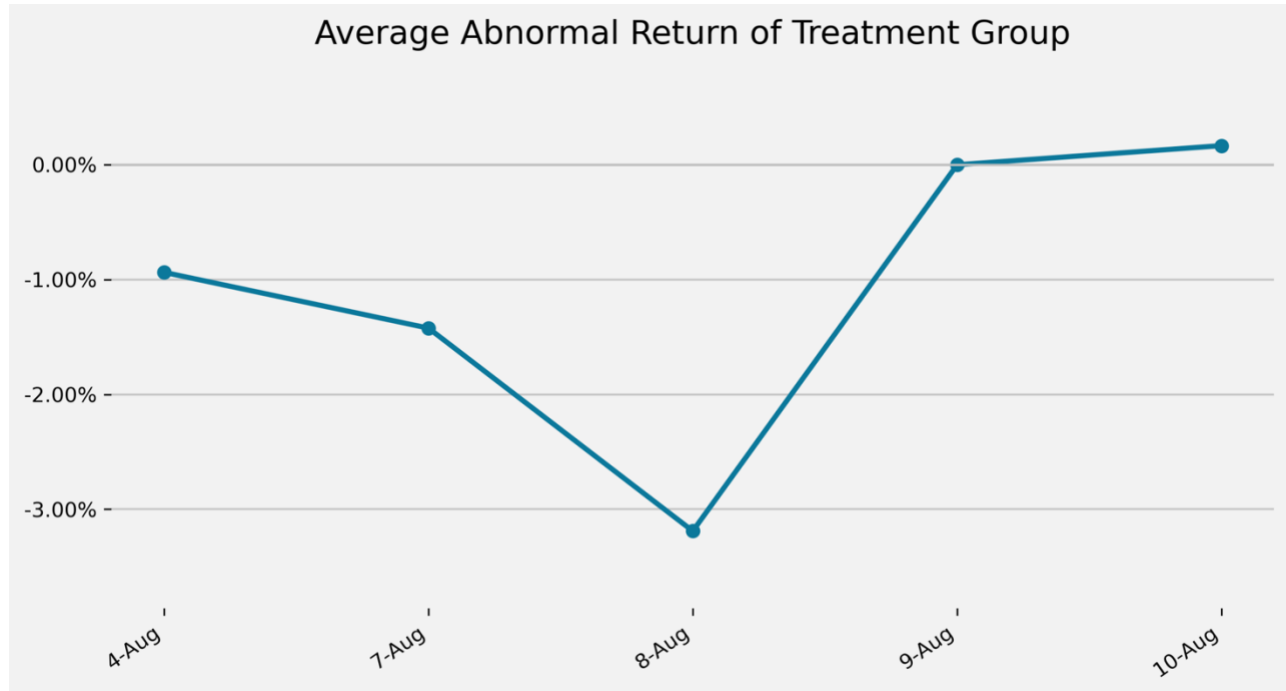


Figure 1: Average Abnormal Return of Treatment Group

Figure 1 shows the AAR of the treatment group. Before the event day, the AAR was already negative, falling from -0.94% on August 4 to -1.42% on August 7. This implies that the market

might have expected bad news in advance, causing a decline. On the event day (August 8), the AAR sharply fell to -3.19%. Many companies in the treatment group are in business closely related to obesity, such as weight-loss surgery and breathing machines. They have direct competition with GLP-1 drugs. When investors heard the SELECT trial results, they expected that these companies' future profits would be greatly reduced, so they chose to sell their shares. After the event day, the AAR quickly returned to near zero. This shows that the panic selling was mostly concentrated on the event day. The market absorbed the bad news within one day, and there was no continued abnormal drop afterwards.

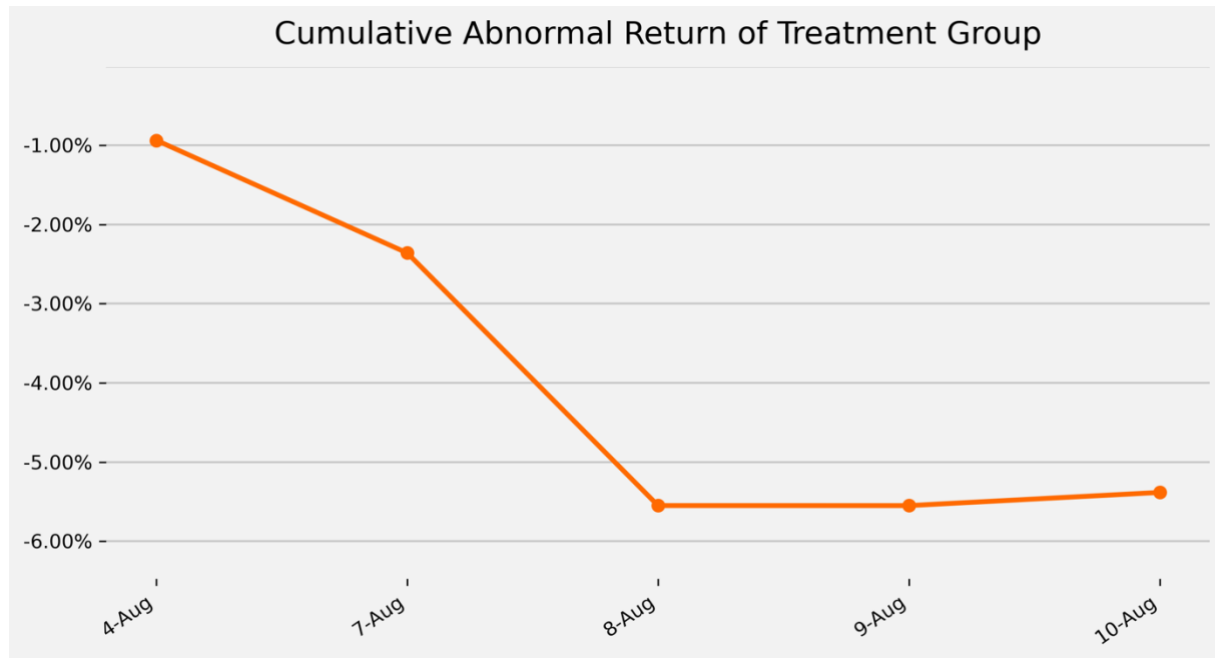


Figure 2: Cumulative Abnormal Return of Treatment Group

Figure 2 illustrates the CAR of the treatment group. Before the event day, negative returns kept adding up, reaching -2.35% on August 7. On the event day, the CAR dropped to -5.46%. After the event day, the CAR stayed around -5.5% to -5.3% and did not go back up. This means the event moved the stock prices of the treatment group to a new and lower average level.

3.3 Control Group

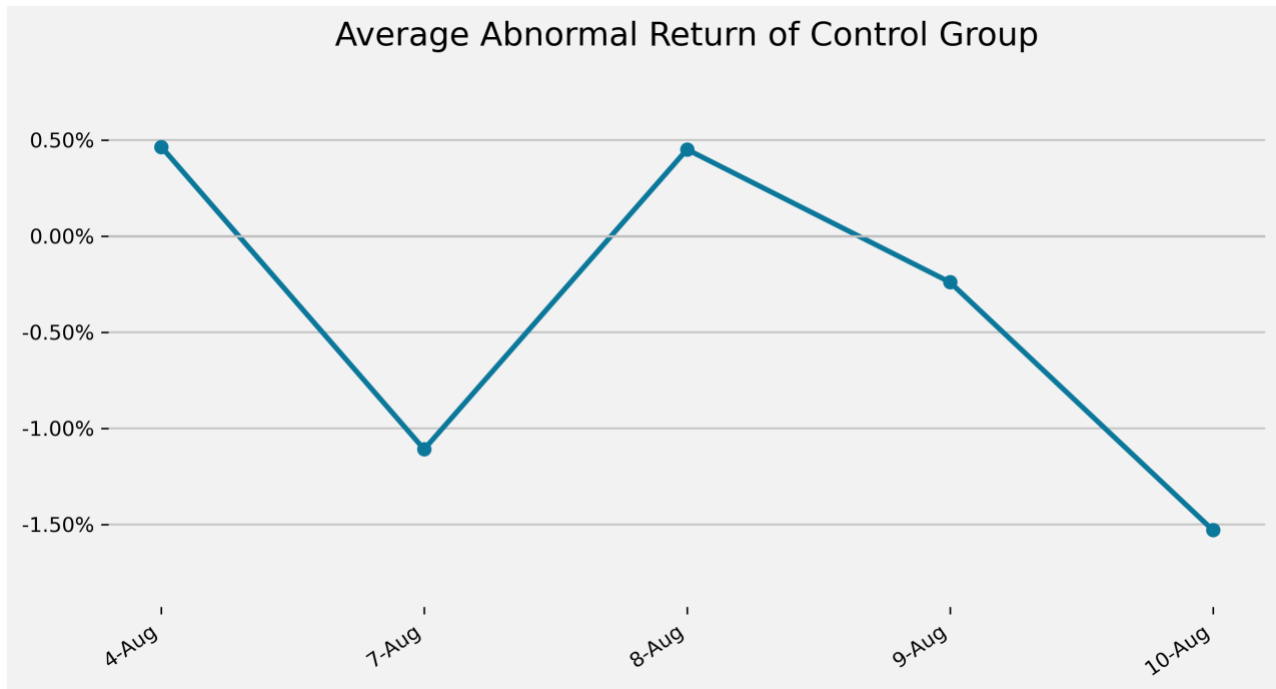


Figure 3: Average Abnormal Return of Control Group

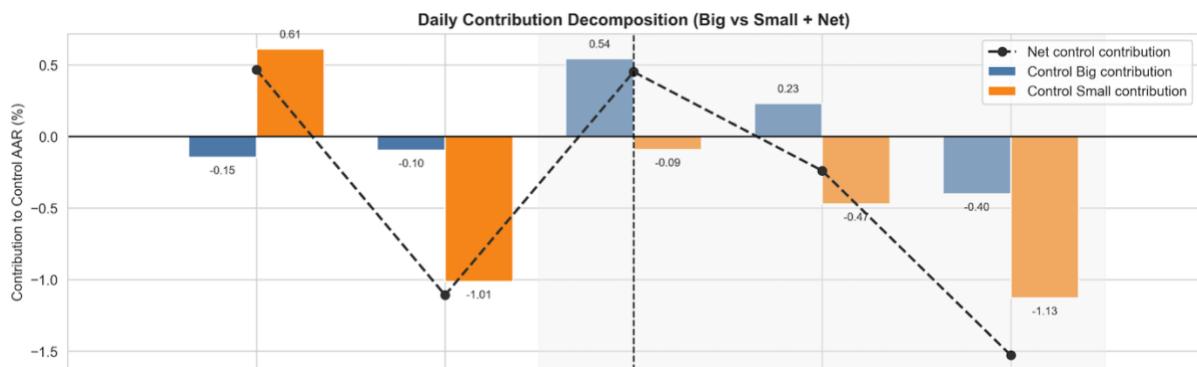


Figure 4: Daily Contribution Decomposition (Big vs Small)

Figure 3 shows the AAR of the Control group. Before the event day, the AAR of the control group went down, from 0.47% on August 4 to -1.11% on August 7. On the event day, the AAR of the control group rose to a positive number (+0.45%). This shows that the bad news about GLP-1 drugs did not hurt the companies in the control group. Instead, a small amount of money flowed into them. This may be because investors sold treatment group stocks and bought unaffected companies. But after the event day, the AAR of the control group fell again, reaching -1.53% on August 10. According to Figure 4, this drop is mainly caused by the small-cap stocks in the control group. Their drop was too big, and they pulled down the average of the whole control group.

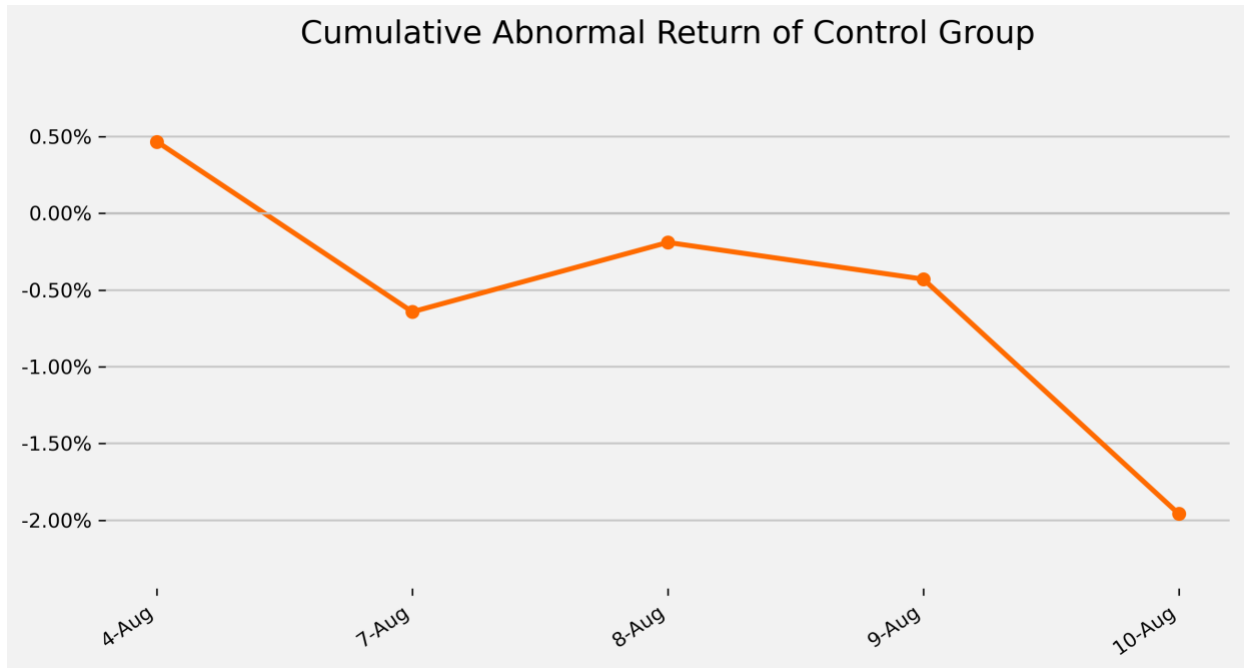


Figure 5: Cumulative Abnormal Return of Control Group

Figure 5 shows the CAR of the control group. It rose slightly on the event day, from -0.65% to -0.20% and then dropped to -1.96% by August 10. Overall, the control group's CAR remains much closer to zero than the Treatment group's CAR.

3.4 Comparison of Both Groups

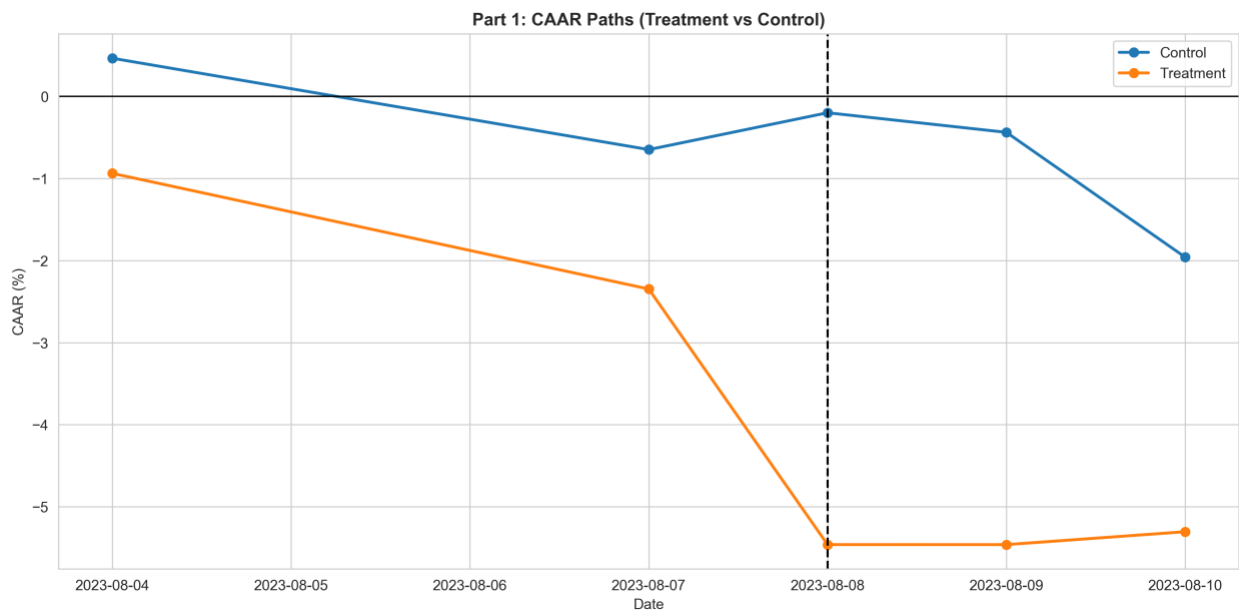


Figure 6: CAAR of Treatment and Control

The figure compares the CAAR paths of the treatment and control groups. Before the event day, both groups showed a downward trend, but the treatment group's decline was greater. On the event day, the CAAR of the Treatment group plummeted to -5.4%, while the CAAR of the control group remained almost unmoved. After the event day, the CAAR of the treatment group stayed at this low level and did not rebound, whereas the CAAR of the control group eventually dropped to -2.0%. This confirms that the event had a severe and persistent negative impact on the treatment group, while the impact on the control group was milder.

3.5. Daily Difference in Average Abnormal Return

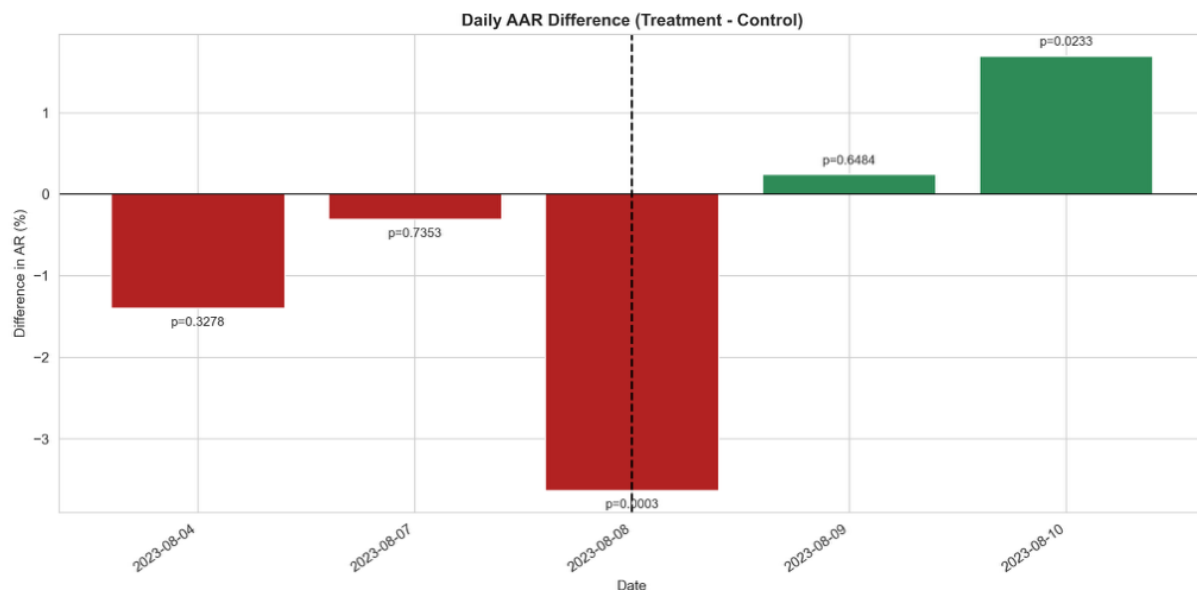


Figure 7: CAAR of Treatment and Control

Figure 7 shows the AAR difference between the treatment group and the control group over the five days. The treatment group underperformed the control group by 3.7% on the day of the event. This was the largest single-day gap in the entire event window. This difference was systematic and not a coincidence, as the p-value is 0.0003. Under the null hypothesis that there is no difference between the two groups, the probability of such a large gap is nearly zero. It can be interpreted that the market processed information rationally and quickly. The results of the SELECT test were clinically significant and quickly understood by algorithmic traders and institutional investors. The gap narrowed to 0.2% one day after the event and was no longer statistically significant, which shows that the intense price changes were concentrated on the day of the incident.

3.6. Best and Worst Performers (ranked by CAR)

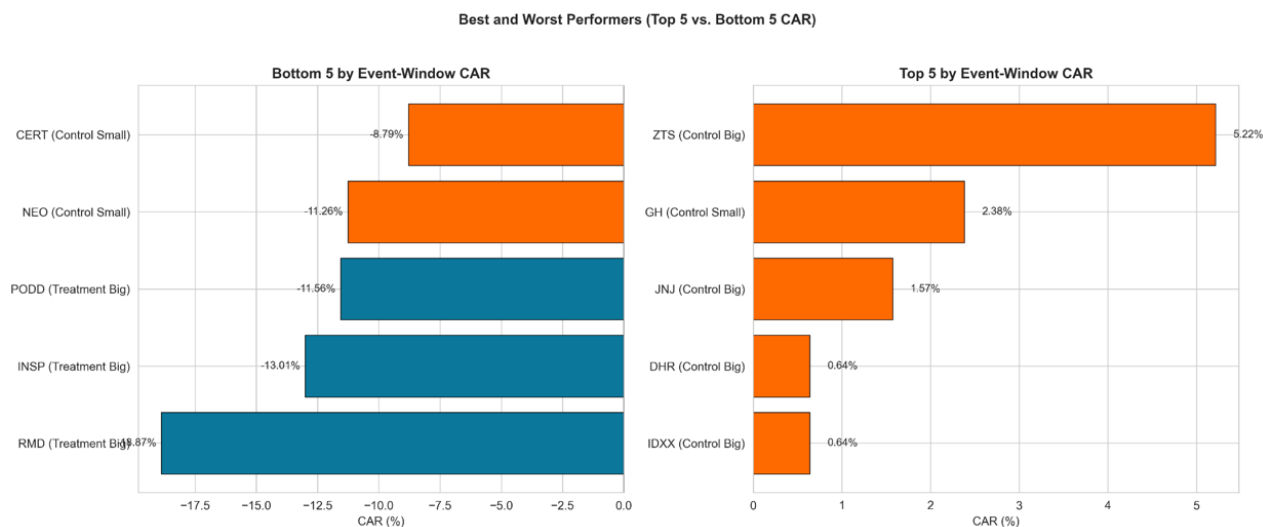


Figure 8: Best and Worst Performers

Figure 8 shows the best and worst performing stocks based on the cumulative abnormal rate of return (CAR) during the five-day event window period. The top five stocks are all from the control group. Their business model is completely independent of human metabolic health, so it is not affected by the negative news of GLP-1. When investors sell the shares of the treatment group, they rotate their funds to these stocks that are regarded as "safe havens", forming an alternative effect.

Of the five worst-performing stocks, three are from the experimental group (RMD, INSP, PODD), whose core business is heavily dependent on obesity-related complications. The SELECT test directly weakens the fundamental demand for its products. However, the other two worst performers came from the control group (CERT, NEO). Both companies' products are irrelevant to obesity or diabetes, but their decline (-8.79% and -11.26% respectively) even exceeds that of some experimental companies.

Why do companies that are completely unrelated to fundamentals also plummet? The reason behind is liquidity and the macro environment. When the SELECT results were announced, some investors did not have time to judge which specific medical stocks would be exposed to the risk. They decided to sell it first and ask questions later. Small-cap healthcare stocks such as CERT and NEO have a small trading volume, there are insufficient buyers to absorb the selling pressure, causing the stock price to plummet. Investors rush to dump these small-cap stocks first because they believe that if they wait until later, they might get stuck without a buyer.

The macro environment further amplifies this liquidity-driven price shock. The event occurred at a time when the market expected the Federal Reserve to raise interest rates aggressively (TRADING ECONOMICS, n.d.). Small-cap stocks have already lost the favor of investors due to the lack of profitability. The GLP-1 negative news hit when small-caps were already weak, which

made them go further lower. The collapse of CERT and NEO reveals a key insight: fundamental innocence is not enough to protect a stock from selling pressure in a stressful event.

3.7. Core Findings of Event Study

Three core findings were derived from the event study, which provided a basis for the subsequent long-term analysis. First, most market reactions are concentrated on the event day, with a small negative AAR two days before the event indicating potential information leakage. Second, the market shows high accuracy in distinguishing between exposed and unexposed companies. With a p-value of 0.0003, the large gap of -3.7% between the two groups on the event day indicates a systematic differentiation rather than random selling. Lastly, it can be observed that it is a structural revaluation instead of a temporary panic since the CAR curve did not rebound after the incident. Overall, these findings demonstrate that the market's response to the SELECT trial announcement was rapid, accurate and lasting.

4. Individual Stock Analysis

4.1 Average return

The comparison of the average returns of the two groups before and after the event showed obvious differences (Figure 9). The overall return of the experimental group fell significantly, and many stocks changed from positive to negative, highlighting the serious market shock suffered by the group. In contrast, although the control group also experienced fluctuations, its income distribution was relatively stable, avoiding the systematic collapse experienced by the experimental group.

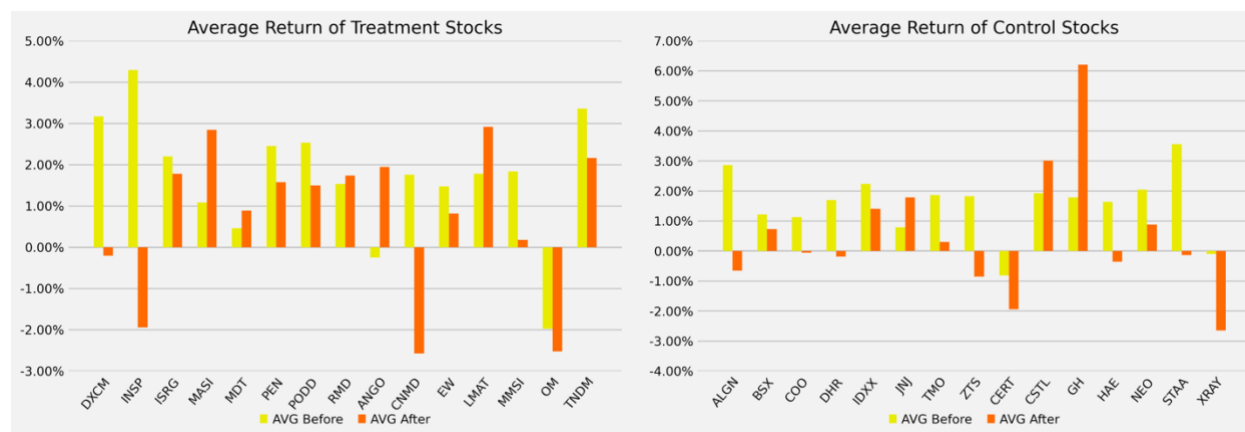


Figure 9: Average Stock Returns Before and After the Event

In the experimental group, the performance of INSP and CNMD was the most prominent. Both stocks made considerable gains before the incident - the yield of INSP was close to 4.5% - but plummeted to negative value after the incident, and CNMD fell by more than 2.5%. This drastic reversal shows that the market has quickly adjusted its expectations for the future cash flow of these companies, effectively repricing them from "high-growth/high-premium" assets to "high-risk/low-return" securities. OM's performance was particularly extreme; the stock was already in

the negative range before the incident. After the incident, the decline intensified, indicating that the incident worsened the original fundamental weakness and triggered a sell-off wave. On the contrary, stocks such as MASI rebounded. This resilience can be attributed to MASI's core business-non-invasive testing technology, which has been deeply integrated into the workflow of clinical hospitals (Goldman, Petterson, Kopotic & Barker, 2000). The high conversion cost of its products forms an "immune mechanism" that allows MASI to maintain its risk-averse position in the face of difficulties in the whole industry. In addition, MDT and ANGO had lower yields before the incident, but both achieved growth after the incident. This "low opening and high closing" model indicates that the market may have reassessed these undervalued assets after the event, or investors believe that the beta coefficient of these assets is low, so they are held as defensive assets during the spread of panic.

Although the control group is theoretically not directly affected by the event, the data shows that there is a significant difference in their performance, indicating a spillover effect. GH performed the most outstandingly, with an explosive increase in yields after the incident, from less than 2.00% to more than 6.00%. This excellent performance may stem from the Substitution Effect": as funds flow out of the experimental group, it flows into the leading enterprises of the control group with similar or highly relevant business models, and GH becomes the main beneficiary. CSTL has also shown strong growth momentum, with a significant increase in returns after the incident, indicating that its stable financial situation has successfully attracted new capital seeking risk avoidance. However, not all stocks in the control group survived; the stock prices of XRAY and CERT fell sharply, with the yield of XRAY falling to about -2.5%. This shows that the control group is not an iron-paneled haven. This phenomenon reflects that when an event occurs, the market often sells the stocks of the whole industry indiscriminably (Irwin, 1994), resulting in even fundamentally stable stocks being squeezed by valuation.

4.2 Beta

After the incident, Figure 10 shows that the Beta value of the two groups of stocks has generally risen, reflecting the market's reassessment of risk premiums to cope with sudden shocks. The overall increase in Beta value indicates that the ability of individual stocks to decouple from the overall market trend is weakening. In a highly uncertain environment, investors tend to give up focusing on the fundamentals of the company and choose to hedge or liquidate through macro

indices such as **NASDAQ or S&P 500**. This behavior has manically increased the correlation between individual stocks and the overall market.

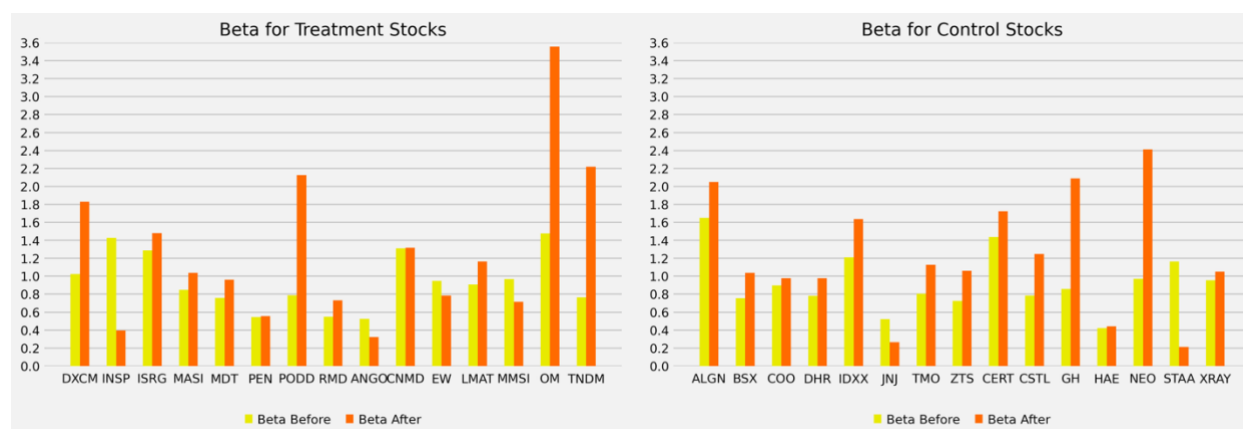


Figure 10: Changes in Stock Beta Before and After the Event

In the experimental group, the sharp rise in the Beta value of OM and PODD stocks revealed the emotional pricing of these assets in the market. When the Beta value reaches such a high level, it shows that the volatility of these stocks is almost completely influenced by macroeconomic sentiment. For investors, this means that these assets have lost their defensive role and turned into pure exposure tools. Any slight fluctuations in the market will be nearly tripled by these stocks, and it is easy to trigger chain selling through algorithmic trading. In stark contrast to this, the risk stability of DXCM is higher. Although its Beta value has increased, it is still within a relatively controllable range, indicating that the market consensus on the fundamentals of DXCM is still solid. Even in the case of market fluctuations, institutional investors still trade based on business logic rather than panic, avoiding becoming blind followers of market fluctuations.

The Beta value of the control group is not uniform; the Beta value of NEO jumped from about 1.0 to more than 2.4, showing extreme abnormalities. A Beta value close to 1.0 usually indicates that its performance is consistent with the market benchmark, while the Beta value of 2.4 means that its sensitivity suddenly soars. This shows that after the incident, the market reclassified NEO, which may reveal the hidden links between its business operations and the affected treatment group - such as supply chain dependence or shared financing channels - thus transforming it from a stable investment to a high-risk asset. It is worth noting that both INSP and ANGO's beta values have decreased after the incident. This phenomenon may reflect flight-to-quality, that is, investors redeploy funds to assets that are considered to be fundamentally stable or have a defensive business model during periods of violent market fluctuations. Under systemic pressure, risk-averse assets usually show a negative or significantly weakened correlation with the overall market, indicating their relative independence (Baur & Lucey, 2010). In addition, the decline in the Beta value may stem from its idiosyncratic resilience, which means that the company's business logic is highly isolated from the specific event. According to the theory of heterogeneous risk (Roll, 1988), when a company's cash flow is mainly driven by specific factors rather than macroeconomic conditions, its sensitivity to overall market fluctuations will naturally decrease during the crisis.

4.3 SML

Before the event, the securities market line (SML) presented a standard positive slope (Figure 11), which is in line with the classic capital asset pricing model (CAPM). In this model, investors who bear higher systemic risks (Beta value) will receive corresponding risk premium compensation. On the contrary, after the incident, the slope of SML tends to be significantly flat, and even shows a negative slope in some high Beta values, while the discreteness of individual asset returns also increases significantly. This slope reversal indicates that the CAPM pricing framework fails under extreme market conditions, which is similar to the risk premium inversion. In this case, high-beta stocks not only fail to produce excellent risk compensation, but also their returns are far lower than the market average due to panic selling. Investors no longer allocate assets based on risk-return trade-offs, but sell indiscriminately out of liquidity needs, which leads to the collapse of the systematic risk compensation mechanism.

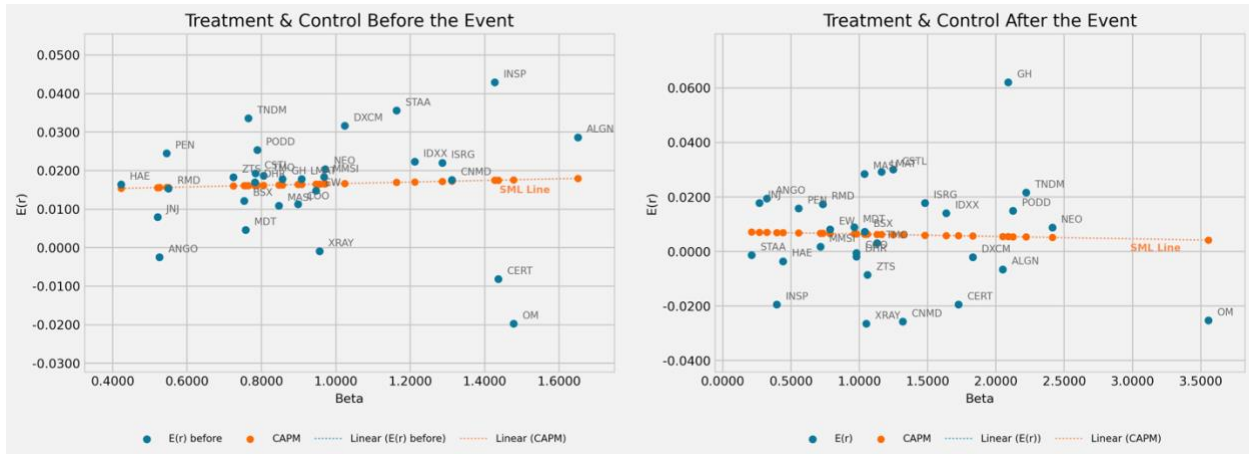


Figure 11: SML Analysis by Group

As shown in Figure 12, the comparison between the processing group and the control group reveals significant differences in pricing distortion. For the processing group, the slope of the SML is negative, and the R^2 value changes from -0.134401 to -0.234952. This shows that the processing team experienced intense panic after the incident. Investors' risk expectations of these assets are completely out of the fundamentals, because the market regards them as high-risk, low-value and liquidated assets. In this case, if a higher systemic risk directly leads to greater losses, it means that the market pricing logic has completely collapsed under extreme pressure.

In contrast, although the SML slope of the control group tended to be flat after the incident, its R^2 value was still positive (0.046487). Despite the deterioration of the market environment, the pricing logic of the control group did not completely collapse. A positive R^2 value indicates that there is still a weak but continuous correlation between the return and risk level of the group, indicating that the control group showed stronger pricing resilience. The market continues to trade these assets at the benchmark risk assessment level, further confirming that the treatment group has suffered more serious market deviations or liquidity shocks.

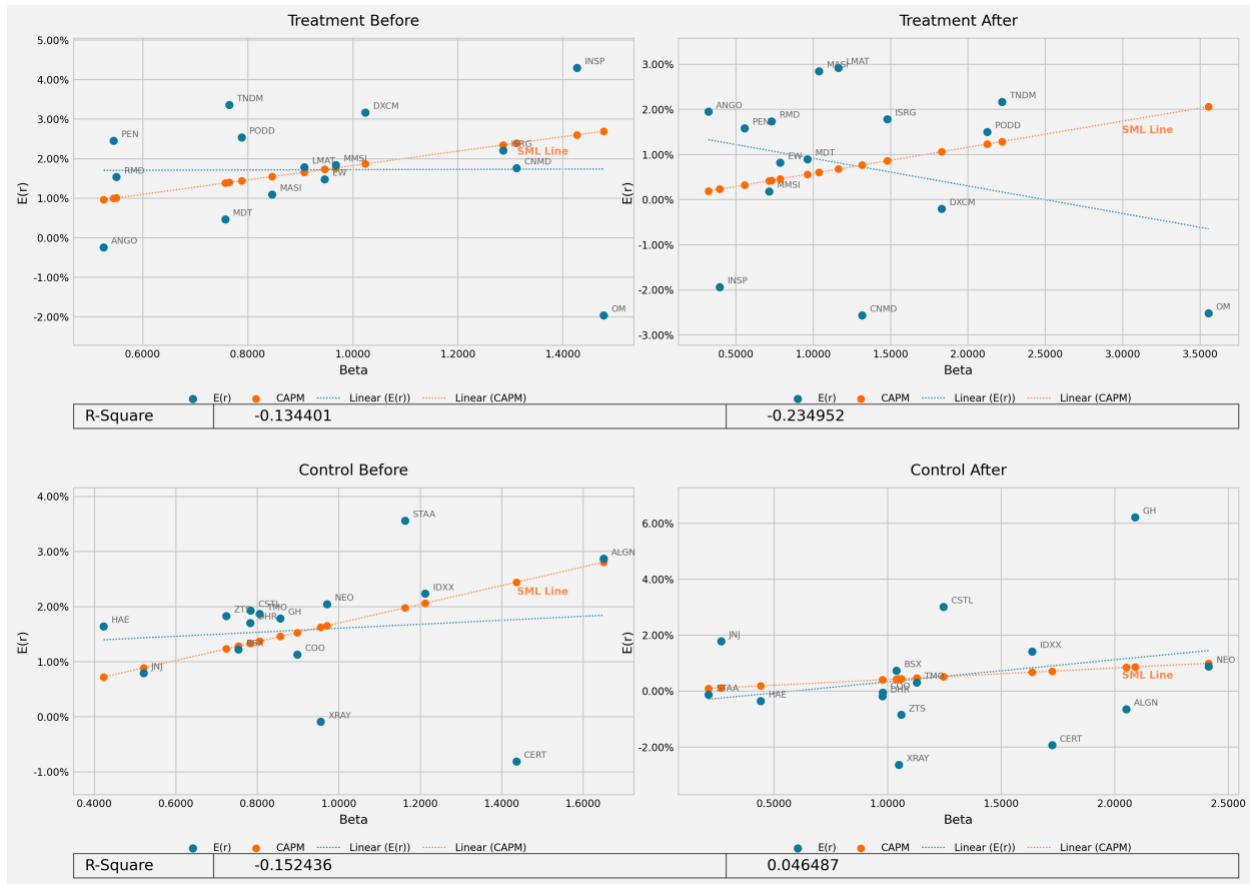


Figure 12: SML Regression Statista at Individual Stock Level

5. Firm Size Analysis

This section will further explore whether the company size affects the impact of GLP-1 event on stock risk and return. This report divides the treatment group and the control group into two subgroups based on the market capitalization of each company: Big (representing large market capitalization) and Small (representing small market capitalization). The specific grouping is shown in Table 3:

Treatment Group			Control Group		
Ticker	Name	Group	Ticker	Name	Group
ISRG	Intuitive Surgical	Treatment Big	JNJ	Johnson & Johnson	Control Big
DXCM	DexCom	Treatment Big	TMO	Thermo Fisher Scientific	Control Big
RMD	ResMed	Treatment Big	DHR	Danaher	Control Big
PODD	Insulet	Treatment Big	ZTS	Zoetis	Control Big
MASI	Masimo	Treatment Big	IDXX	IDEXX Laboratories	Control Big
MDT	Medtronic	Treatment Big	COO	Cooper Companies	Control Big
PEN	Penumbra	Treatment Big	ALGN	Align Technology	Control Big

INSP	Inspire Medical Systems	Treatment Big	BSX	Boston Scientific	Control Big
TNDM	Tandem Diabetes Care	Treatment Small	XRAY	Dentsply Sirona	Control Small
OM	Outset Medical	Treatment Small	HAE	Haemonetics	Control Small
CNMD	CONMED	Treatment Small	CERT	Certara	Control Small
EW	Edwards Lifesciences	Treatment Small	NEO	Neo Genomics	Control Small
MMSI	Merit Medical Systems	Treatment Small	CSTL	Castle Biosciences	Control Small
LMAT	LeMaitre Vascular	Treatment Small	GH	Guardant Health	Control Small
ANGO	AngioDynamics	Treatment Small	STAA	STAAR Surgical	Control Small

Table 3: Study sample grouping based on market capitalization

- Treatment Big: These enterprises hold a dominant position in their respective niche markets and have strong customer loyalty.
- Treatment Small: These companies have relatively narrow business scopes and are more dependent on specific products or indications. Therefore, theoretically, they are more vulnerable to the risk of GLP-1 substitution.
- Control Big: These enterprises have highly diversified business operations. The impact of GLP-1 drugs on their core businesses such as pharmaceuticals and cardiovascular devices is limited.
- Control Small: These companies have relatively low correlations with obesity or metabolic diseases, thus serving as a benchmark for low-risk and small-cap stocks.

Through the above four divisions, this paper will conduct comparative analyses respectively based on the average return rate, the security market line (SML), and the Sharpe ratio.

6. Return Analysis

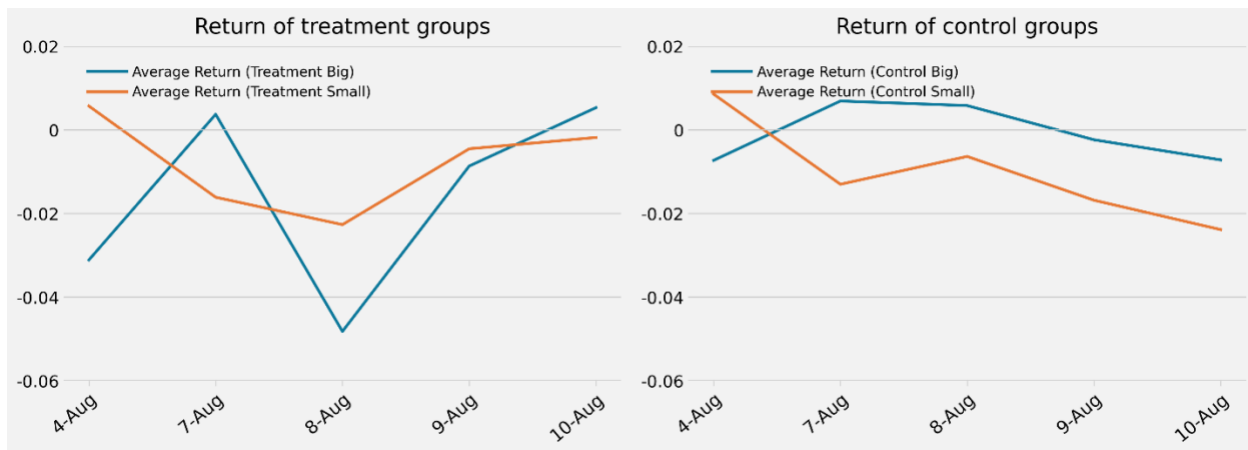


Figure 13: Average return of four groups

Figure 13 shows daily average returns of 4 groups from August 4 to August 10, 2023. Both large-cap and small-cap treatment stocks experienced sharp negative returns on the event day. The large-cap stocks suffered a greater decline, reaching approximately -5%, and also had higher volatility. This indicates that larger and more developed medical device companies are more sensitive to the threat of GLP-1 drug substitution and respond more quickly to new information. It is notable that the decline in the Treatment Large group was twice that of the Treatment Small group, suggesting that large-cap medical device companies face greater valuation adjustments. The possible reason is that these large companies had higher market expectations before the event, so there was a greater potential for their valuations to be reduced after the results of the SELECT trial were announced.

For control group, control stocks show much smaller fluctuations overall. Large-cap control firms remain relatively stable. Small-cap control firms show a gradual downward trend, but without an event-day crash.

The impact of the event on the treatment group was more significant than that on the control group. This phenomenon may indicate that large-cap stocks with high exposure have better liquidity and more concentrated institutional holdings, thus being sold off more quickly under information shock. For the stocks in the treatment group, the impact of the event dominated over the company size effects. But for the stocks in the control group, larger companies were more stable.

7. Return and Beta Analysis

The security market line (SML) illustrates the relationship between the expected return rate of investment and its risk (Kenton, 2025). This part presents SML analyses for each group before and after the event.

7.1 Treatment Big Group

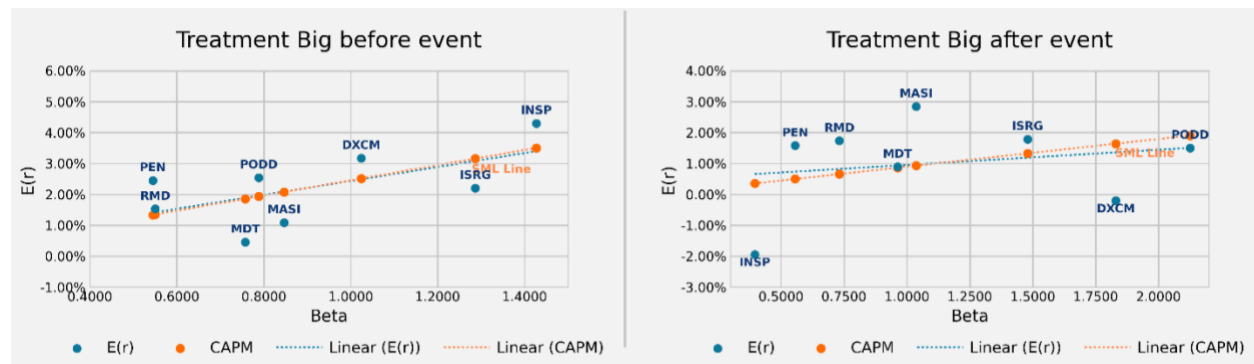


Figure 14: SML – Treatment Big Group

Before the event, Figure 14 shows that SML was upward-sloping and presented a positive risk-return relationship. Stocks with higher beta coefficients were expected to have higher returns, which was in line with the CAPM theory's expectation of high returns for high risks. However,

some stocks such as DXCM and PODD deviated from the SML line, indicating that their returns were greatly influenced by specific company factors. Therefore, the risk-return relationship of this group of stocks was established in the market, but there were some pricing differences among individual stocks.

After the event, the SML of the Treatment Big group still maintained a positive slope, and the risk-return relationship was still effective. However, the actual expected rate of return of most stocks is above the CAPM forecast line, indicating that the actual return of these stocks exceeds the theoretical return that should be obtained based on their Beta, meaning they have a positive alpha. This phenomenon may reflect the market's reaction to the GLP-1 event may be excessive selling, resulting in the undervaluation of some companies that are still stable in fundamentals. In addition, investors give positive expectations for the long-term adaptability or business diversification of these companies.

However, DXCM and INSP are significantly lower than the SML line and present negative alpha, reflecting the market's stronger concern about the threat of GLP-1 drug substitution in the products of the two companies, believing that their actual returns are difficult to compensate for the systemic risks. This differentiation shows that even if they belong to the same high-cap group, there is a significant difference in the market's judgment of the degree to which different business lines are impacted by GLP-1.

7.2 Treatment Small Group

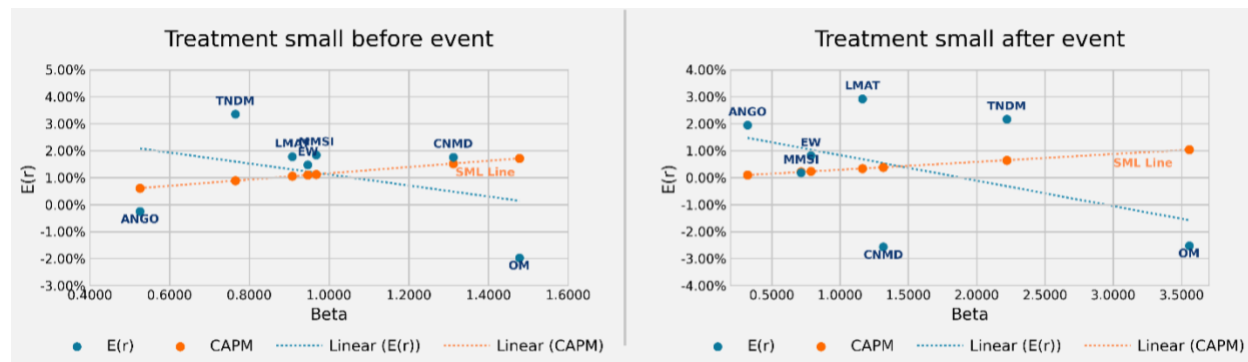


Figure 15: SML – Treatment Small Group

Before the event, based on Figure 15, the SML of the Treatment Small group showed an upward positive correlation, which was in line with the theoretical pricing of systemic risks and returns. However, in fact, the fitting line of the actual rate of return of the sample stock shows a downward trend, which is contrary to theoretical expectations. It shows that in the sample range before the event, high beta stocks did not obtain higher returns, but showed returns declined as beta rose.

At the individual stock level, there is a significant deviation between the actual income of different stocks and the theoretical income of CAPM. The real rate of return of TNDM is significantly higher than the theoretical value, and there is a significant positive alpha, while the real rate of

return of OM and ANGO is significantly lower than the theoretical value, showing negative alpha. Although the actual income of most stocks does not deviate much from the theoretical value, the overall explanation of beta to income is limited, and the specific factors of the company have a greater impact on the stock price.

After the event, the phenomenon was further aggravated. Beta generally rose, but stocks with high beta showed greater negative returns, and many stocks fell below SML, showing obvious negative alpha. This shows that after the industry shock, the market's risk assessment of such companies tends to be pessimistic, high systemic risks are no longer compensated by profits, and small-market treatment companies face higher pricing pressure under the impact of events.

7.3 Control Big Group

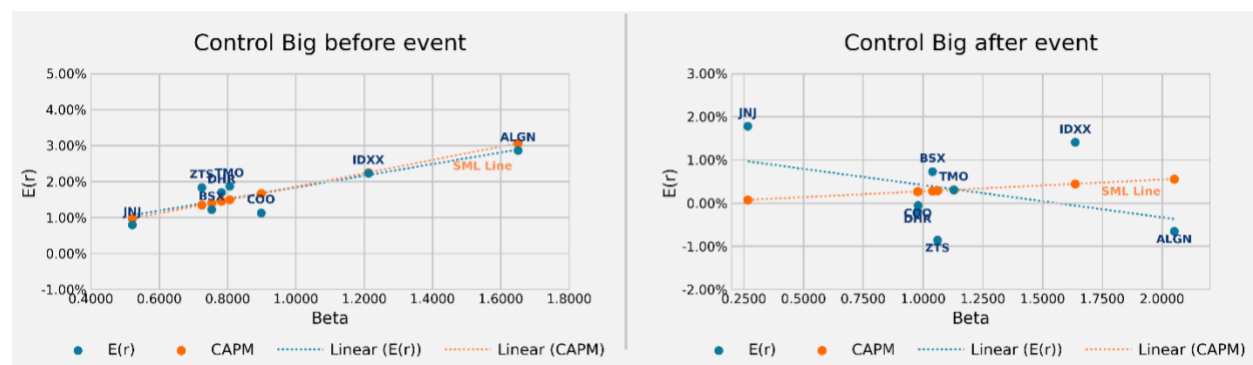


Figure 16: SML – Control Big Group

Before the event, as shown in Figure 16, the actual income fitting line of the Control Big group highly coincided with the CAPM theoretical SML line, and the expected return usually increased with the increase of the beta coefficient. This shows that the risk-return relationship of this group of stocks is reasonable, and the market's pricing of low-exposure large-cap stocks is relatively effective.

After the event, the linear fitting line of the real rate of return deviated significantly from the SML line, showing a downward trend, and CAPM's explanatory power on returns decreased significantly. The differentiation at the individual stock level is intensifying. JNJ was significantly higher than the fitting line, presenting positive alpha, reflecting that its defensive advantage as a diversified health care giant was favored by the market after the incident. In contrast, ALGN and ZTS were located below the SML line, showing negative alpha. This indicates that the market had some concerns about the indirect effect of GLP-1 for its orthodontics and animal health care business. Overall, the market prefers low-beta defense stocks after the event, while high-beta stocks, even those in the control group, also face earnings pressure due to the reprice of business exposure.

7.4 Control Small Group

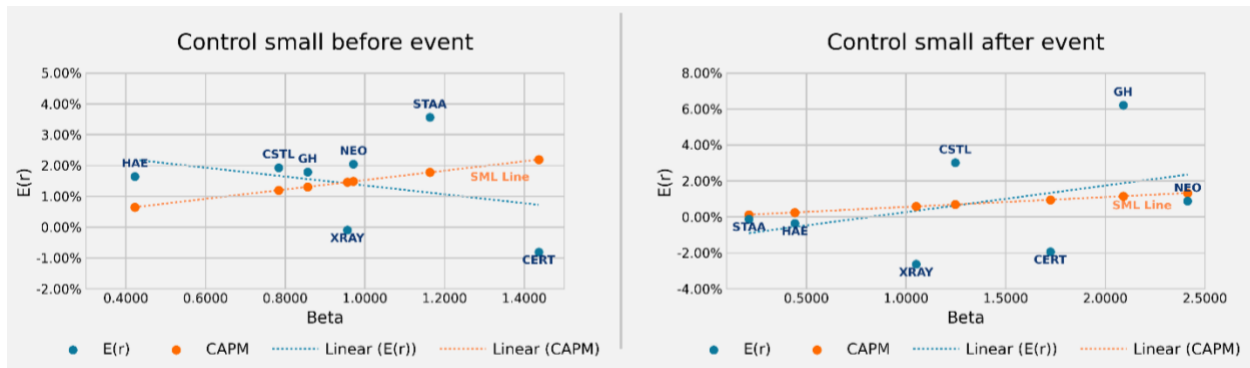


Figure 17: SML – Control Small Group

Before the event, the SML of the Control Small group showed a positive slope, but there was a significant deviation between the actual income and the CAPM forecast line, according to Figure 17. However, after the event, the coincidence between the real rate of return and SML has increased significantly, the market beta coefficient has a stronger explanatory power on the rate of return, and the risk-return relationship tends to be reasonable. However, there is a significant differentiation within the group. GH and CSTL are significantly higher than the fitting line and present positive alpha. Because of the low correlation between its business and GLP-1 drugs, it reflects that the market has given positive expectations for the growth potential of these two genetic testing companies. However, XRAY is significantly lower than the SML line and presents negative alpha, indicating that the market is concerned about the indirect impact of its dental equipment business on the indirect impact of GLP-1 drugs.

Overall, the small control group was less directly affected by the event, and the pricing efficiency after the event was improved largely, which may be related to the rotation of funds from the impacted medical device stocks to such risk-averse targets.

8. Risk-Return Analysis

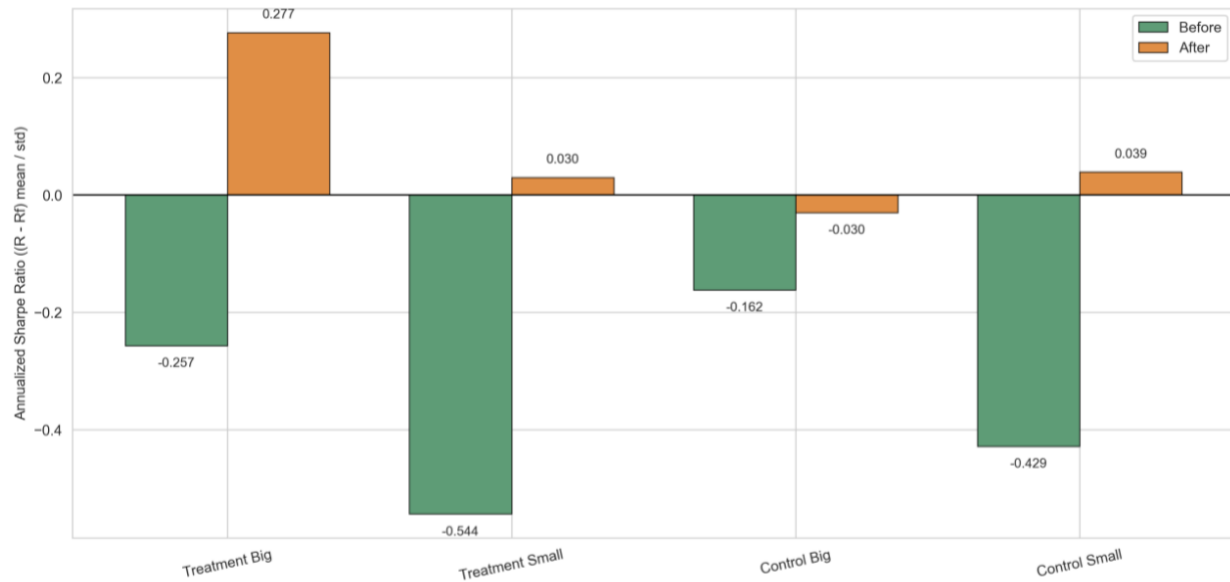


Figure 18: Sharpe ratio of four groups

Figure 18 presents the Sharpe ratios before and after the event for all four groups. The Sharpe ratio measures the excess return obtained per unit of total risk, and is an important indicator for evaluating the risk-adjusted performance of an investment portfolio (Fernando, 2025).

Before the event occurred, the Sharpe ratios of all four groups were negative, indicating that their average returns were lower than the risk-free rate. This suggests that the investment performance was worse than that of risk-free assets, typically implying investment losses and high volatility. Among these four groups, the Sharpe ratios of Treatment Big and Treatment Small were -0.257 and -0.544 respectively, indicating that these high exposure stocks had high volatility but failed to generate corresponding returns before the event. This might be related to the impact of events such as the aggressive interest rate hike cycle of the Federal Reserve during the sample period.

Group	Sharpe ratio
Control - Big	-0.162
Treatment - Big	-0.257
Control - Small	-0.429
Treatment - Small	-0.544

Table 4: Sharpe ratio ranking from best to worst before the event

From the ranking in Table 4, it can be seen that before the event, both the treatment group and the control group had a better risk-adjusted return performance for large-cap stocks compared to small-

cap company stocks. Among them, the Control Small was the group with the worst performance before the event, with a Sharpe ratio of -0.544.

Group	Sharpe Ratio (Before)	Sharpe Ratio (After)	Changes in Sharpe ratio
Control - Small	-0.257	+0.277	↑ 0.534
Treatment - Small	-0.544	+0.030	↑ 0.574
Treatment - Big	-0.162	-0.030	↑ 0.132
Control - Big	-0.429	+0.039	↑ 0.468

Table 5: Changes in Sharpe Ratio For 4 Groups

However, the situation changed after the event occurred. The results presented an overall positive picture. Except for the Control Large, the risk-adjusted performance of the other three groups all turned positive and showed improved risk-adjusted performance. According to the data in Table 5, especially for the small market capitalization group, the Sharpe ratio has significantly improved. This indicates that the impact of events on risk-adjusted returns is asymmetrical.

Furthermore, in terms of risk-adjusted returns after the event, the treatment group outperformed the control group. The Sharpe ratio of the Treatment Big group soared to 0.277, while the Treatment Small group recovered from -0.544 to 0.030. This was because the Treatment group experienced a significant decline on the event day, and the GLP-1 event eliminated the market's uncertainty about the long-term growth logic of these stocks, resulting in a subsequent decrease in volatility and enabling the risk-adjusted returns to be restored. This contrasts with short-term negative reactions in event studies, suggesting that after long-term market repricing, leading companies in the most impacted industry actually receive higher risk compensation.

Therefore, this raises a question that Why did the Control Large become the only worsening group? The Control Large group includes large non-medical device financial and healthcare companies, such as JNJ, TMO, DHR. These stocks were considered as safe assets by investors before the event, because they have diversified businesses, lower volatility and relatively stable returns and cash flows.

However, after the results of the SELECT trial were announced, three possible situations could lead to a long-term impact on the condition of the Control Large group:

- **Capital rotation**

Investors moved their money from these safe large-cap stocks and shifted to other opportunities, such as the oversold rebound of small-cap stocks after a sustained decline in stock prices due to stock selling.

- **Relative attractiveness declined**

When the entire healthcare sector repriced due to this event, the relative attractiveness of these traditional defensive stocks decreases. As a result, the pre-event risk premium disappeared, leading to a deterioration in the Sharpe ratio.

- **Gradual valuation reset**

Since the study focuses on long-term effects, the data after the event covers approximately 3 years. During this period, the impact of GLP-1 drugs was gradually priced into the market. Large healthcare companies may face continuous valuation reductions, meaning that their profit expectations are being revised. Small-cap stocks may have already completed their valuation reset and have relatively stable subsequent performance.

9. Investment Strategy

Long-term analysis shows that the announcement of the SELECT test has triggered a permanent structural revaluation. The average monthly return of the treatment group fell sharply, the beta value rose, and the slope of the securities market line (SML) was upside down. In determining the rate of return, event-specific exposure related to GLP-1 competition has become more important than standard market risk. Based on these findings, we propose two investment strategies for different risk preferences.

9.1 Risk Averse: High Sharpe ratio & positive alpha

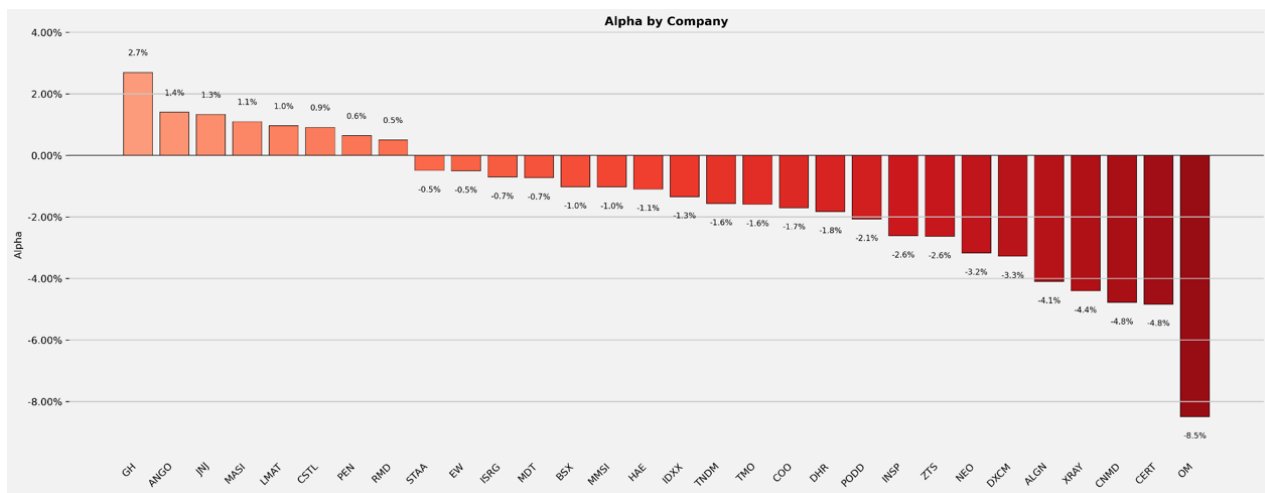


Figure 19: Alpha by Company

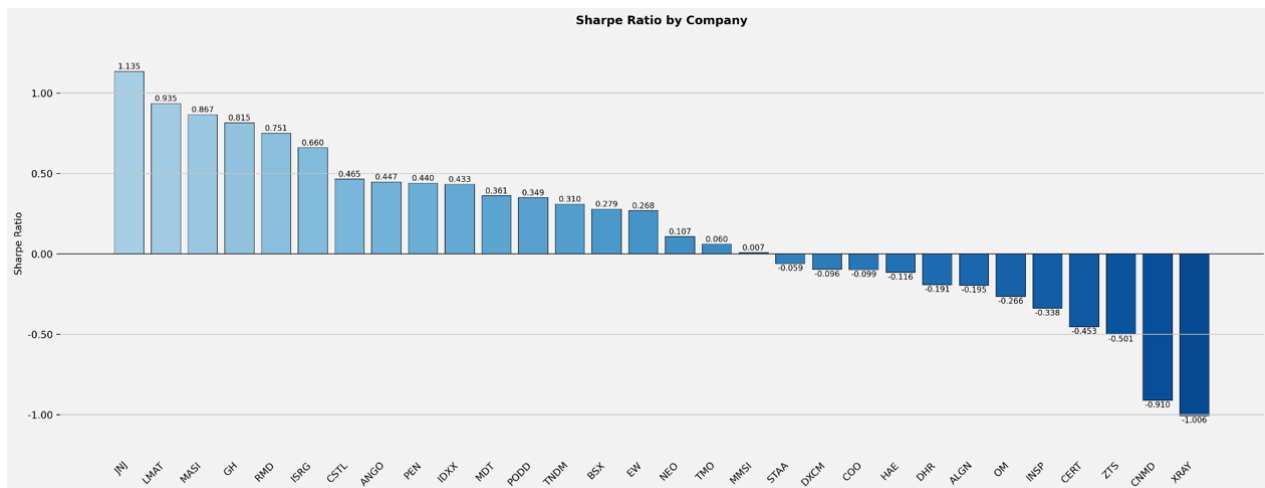


Figure 20: Sharpe Ratio by Company

For risk-averse investors, stability is preferred over high returns. The optimal method is to select the stock with the highest Sharpe ratio and positive alpha value after the event (Figure 19, Figure 20). These stocks are likely to possess superior business qualities such as stable cash flow and resilience to GLP-1 competition, and are expected to continue to generate stable returns that exceed the risk-free rate.

Based on the sorting of the post-event Sharpe ratio and the positive alpha value standard, the following four stocks are recommended: JNJ, GH, CSTL and LMAT. These four stocks all ranked at the top of the Sharpe ratio ranking after the incident and maintained positive alpha values. The strategy involves pure long positions with a long holding period, and the target is to obtain stable returns with lower volatility than the market.

9.2 Risk Lover: Long Treatment Small

For risk-loving investors, high volatility is acceptable in exchange for high expected returns. The appropriate strategy is to focus on making more stocks in the treatment small group, which has the largest improvement in risk-adjusted return after the event.

As shown in Table 4, the absolute value of the Sharpe ratio of small-cap stocks in the treatment group improved the most, from -0.544 before the incident to 0.030 after the incident, with a difference of +0.574, far exceeding any other group. Although the treatment group's large-cap stocks achieved the highest monthly absolute rate of return after the event, the treatment group's small-cap stocks provided the maximum potential for continuous revaluation, because the market was the most pessimistic about the pricing of these stocks before. Moreover, treatment-small has the highest beta value in all groups, providing the volatility sought by risk-loving investors.

Within the treatment small group, stocks should be selected according to the strongest fundamentals after the event (Table 6).

Ticker	mean_monthly_return	std_monthly_return	alpha	beta	reg_r2	mean_monthly_return_pct	alpha_pct	Strategy_Role
ANGO	0.019452873	0.137852859	0.014026725	0.32332495	0.006456714	1.945287318	1.402672454	Alpha Focus
LMAT	0.029183682	0.102008618	0.009669493	1.162781324	0.152506024	2.918368212	0.966949273	Risk-Neutral Core
EW	0.008156423	0.084009826	-0.005034979	0.786028858	0.10274991	0.815642348	-0.503497863	Other
MMSI	0.001782965	0.063660581	-0.010217238	0.715049486	0.148079735	0.178296474	-1.021723755	Other
TNDM	0.021653203	0.22375266	-0.015614177	2.22063095	0.115606166	2.16532028	-1.561417676	High-Beta Satellite
CNMD	-0.025697946	0.104092524	-0.047798445	1.316890326	0.18785587	-2.569794626	-4.779844504	Other
OM	-0.025207118	0.350258645	-0.084877142	3.555525053	0.120946803	-2.520711793	-8.487714175	Other

Table 6: Treatment Small Post Fundamentals

For instance, LMAT is the most stable, with a monthly post-event return of 2.92% and a positive alpha value of 0.97%. ANGO has the highest alpha value in the group, reaching 1.40%, and the post-event return is 1.95%. TNDM provides a high beta value (2.22), with a post-event return of 2.17%. The strategy involves long-only to hold positions without hedging, fully accepting market risk exposure. Due to the high risk, the strategy should be limited to 20% of the total portfolio. The small-cap stocks in the treatment group have the highest beta value among all groups (Table 7), reaching 1.44, which means that their volatility is 44% higher than that of the market.

Group	mean_monthly_r	std_monthly_ret	alpha	beta	reg_r2	mean_monthly_beta	beta_rank_desc
Treatment Small	0.004189155	0.177615681	-0.019977966	1.440032992	0.07500884	0.418915459	1
Control Small	0.007184936	0.189822328	-0.01482147	1.311283668	0.054453888	0.718493551	2
Control Big	0.003105271	0.085614602	-0.01606569	1.142329539	0.203032731	0.310527083	3
Treatment Big	0.01021934	0.122944746	-0.008906893	1.139664422	0.09799704	1.021934024	4

Table 7: Group beta after event

9.3 Risk Neutral: Quality Resilience

For risk-neutral investors, the goal is to balance between risk and return, neither extremely averse to risk nor radically pursuing high returns. The appropriate strategy is to choose those stocks that fell less than similar stocks during the event, recovered quickly and produced a positive alpha after. These stocks possess genuine qualities such as stable cash flow and resilience to the GLP-1 shock, and the initial overreaction of the market proved to be temporary.

There are three selection criteria (Figure 8, Table 7). First, the decline in the AAR on the event date is less than the group average, indicating that investors believe that it's less exposed to risk or the business fundamentals are stronger. Second, the CAR recovered rapidly after the event, suggesting that the market recognized the overreaction and made corrections. Third, the positive alpha post-event confirms that the stock provides a return beyond what the market explains.

Based on these criteria, LMAT is primarily recommended. The AAR decline on the event day for LMAT was only -0.43%, far better than the treatment small group average of -1.83%. Its CAR recovered to -0.65% at the end of the event window, and produced a positive alpha value of 0.97% per month after the event (Table 6). The strategy aims to capture the correction of the market's initial overreaction and avoid extreme volatility with long-only positions and a long holding period.

10. Conclusion

Overall, our evidence shows that the SELECT trial announcement on August 8, 2023, triggered a clear and immediate cross-sectional repricing within the healthcare universe, consistent with a “demand reallocation” narrative in which GLP-1 therapies are expected to substitute for parts of the obesity-related device and procedure ecosystem. On the event day, the treatment group experienced a sharply negative abnormal return ($AAR = -3.19\%$), while the control group remained positive ($AAR = +0.45\%$), producing a statistically significant gap (difference $\approx -3.64\%$, $p < 0.01$). Importantly, the strongest reaction was concentrated on the event day rather than gradually accumulating across the full window, supporting the interpretation that the market processed the new information rapidly.

Beyond the short-horizon shock, the price response appears structural rather than temporary. The treatment group’s cumulative abnormal performance dropped to roughly -5.5% on the event day and then remained near that lower level without a meaningful rebound, implying a persistent valuation reset instead of a short-lived liquidity-driven overshoot. The long-term diagnostics reinforce this interpretation: risk measures (betas) generally increased after the event, and the relationship between beta and return (captured by SML/CAPM patterns) weakened or even inverted for the most exposed firms, suggesting that event-specific (idiosyncratic) exposure to GLP-1 disruption dominated standard risk pricing. In addition, the size split indicates that exposure dominated firm size for treatment stocks, while larger firms in the control group tended to be more stable highlighting that “who is exposed” mattered more than “how big the firm is” when the shock is fundamentally technological/therapeutic.

These findings carry direct investment implications. For conservative investors, a market-neutral relative-value approach is more appropriate than broad directional bets: hedged long/short pair trades can short structurally vulnerable obesity-linked device names while going long more resilient, diversified healthcare firms to reduce sector beta and isolate disruption exposure. For more aggressive risk-taking, selective contrarian opportunities may exist when the market initially overreacts to firms whose products are complementary rather than substitutable to GLP-1 adoption, but such trades require disciplined risk controls (e.g., strict stop-loss rules). Across both styles, the overarching discipline is to avoid a blanket “buy-the-dip” mentality in the most exposed names, because the evidence is more consistent with a permanent fundamental re-rating than a reversible panic sell-off.

Finally, the conclusions should be interpreted with standard empirical limitations in mind. Results may depend on the specific stock sample, group classification, and chosen event window, and future work could strengthen external validity by expanding the sample, testing alternative benchmarks/models for abnormal returns, and examining additional GLP-1–related information releases to confirm whether similar repricing patterns persist across events.

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Appendix

https://github.com/dcjznb/PolyU_AF3316_Investment